BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 132 · Lesson 132 · Hybrid Encryption: Protecting a Connection

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Which key verifies a signature produced using an owner's private key?

Q2. Which memory is typically volatile: RAM or ROM?

Q3. In a queue, what does FIFO stand for?

Q4. What is O(n) search complexity associated with?

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

session key — Temporary symmetric key for a connection/session.

hybrid encryption — Combines asymmetric key establishment with symmetric data encryption.

certificate — Credential binding identity to a public key, subject to trust validation.

confidentiality — Preventing unauthorised reading of information.

New learning

Authenticate and agree — Client validates server identity and securely establishes shared key material.

Derive session key — Both ends obtain secret symmetric traffic keys.

Encrypt traffic — Bulk application data travels encrypted under symmetric cipher.

Worked example

Explain why a modern web connection does not encrypt every file byte directly with an asymmetric cipher.

1. Check server certificate — Predict the first step from the prompt.

2. Derive symmetric traffic key — Explain the intermediate result or decision.

3. Encrypt application data — Modern secure connections typically authenticate with public-key credentials, establish shared session keys (often via ephemeral key agreement) and use efficient symmetric encryption for bulk traffic. Identity checks guard against impostors.

We Do · Practise with guidance

A pupil visits a fake website showing a padlock. Does encryption prove the website is trustworthy?

HTTPS is encrypted

Predict the first step from the prompt.

Check actual domain

Explain the intermediate result or decision.

Assess site legitimacy

No. HTTPS can establish an encrypted connection even with a harmful operator. Verify the domain and service, not just the presence of encrypted transport; a certificate

proves control of an identity/domain to varying levels.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Describe why hybrid cryptography uses asymmetric and symmetric methods together. [6]

Question 2

Explain what a session key is and why it is not the same as a website's public key. [6]

Question 3

Evaluate the assertion 'the padlock means any entered data are safe'. [8]

Exit Ticket · Answer before checking

HTTPS guarantees the website operator is honest.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The corresponding trusted public key.

Q2. RAM.

Q3. First in, first out.

Q4. Linear/sequential search in the worst case.

You Do pages · Independent questions

Q1. Key establishment/authentication [3], efficient symmetric bulk encryption [3].

Q2. Temporary symmetric traffic key [2], derived/established for session [2], distinct from asymmetric identity/public-key material [2].

Q3. Encrypted in transit [2], does not certify honesty/safety of destination [2], phishing/endpoint compromise possible [2], domain checks and practical conclusion [2].

Exit Ticket · Model answer

False. Encryption does not guarantee the operator's intentions.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: